

PEDRO R. LÓPEZ-GÓMEZ

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EMPLOYMENT

Assistant professor , CUNEF Universidad Department of Mathematics	Jan. 2026–Present
Postdoctoral researcher , Universidad de Cantabria Department of Mathematics, Statistics and Computer Science	June 2025–Jan. 2026
Predoctoral researcher , Universidad de Cantabria Department of Mathematics, Statistics and Computer Science	Apr. 2022–June 2025
Research assistant , Universidad de Cantabria	June 2021–Mar. 2022

EDUCATION

PhD in Mathematics , Universidad de Cantabria Advisors: Carlos Beltrán and Ujué Etayo	2025
MSc in Mathematics and Computing , Universidad de Cantabria	2021
BSc in Mathematics , Universidad de Cantabria	2020
BSc in Physics , Universidad de Cantabria	2019

PUBLICATIONS

5. *Minimal Riesz and logarithmic energies on the Grassmannian $\text{Gr}_{2,4}$* , U. Etayo and P. R. López-Gómez, *Revista Matemática Iberoamericana* (accepted), 2026.
4. *Points on $\text{SO}(3)$ with low logarithmic energy*, C. Beltrán, F. Carrasco, D. Ferizović, and P. R. López-Gómez, *Constructive Approximation*, 2026.
3. *Asymptotically optimal t -design curves on the Grassmann manifold*, M. Ehler, C. Karner, and P. R. López-Gómez, 15th International Conference on Sampling Theory and Applications (SampTA), 2025.
2. *Measure-preserving mappings from the unit cube to some symmetric spaces*, C. Beltrán, D. Ferizović, and P. R. López-Gómez, *Journal of Approximation Theory*, 308, 106145, 2025.
1. *Low-energy points on the sphere and the real projective plane*, C. Beltrán, U. Etayo, and P. R. López-Gómez, *Journal of Complexity*, 76, 101742, 2023.

PREPRINTS

2. *Riesz 2-energy of the Diamond ensemble*, P. R. López-Gómez, arXiv:2606.25938, 2026. Submitted.
1. *Discrepancy of determinantal point processes on compact, connected two-point homogeneous spaces*, C. Beltrán, U. Etayo, G. Gigante, P. R. López-Gómez, and R. W. Matzke, arXiv:2605.22295, 2026. Submitted.

THESES

4. *Energy minimization and measure-preserving mappings on spheres, projective spaces, and Grassmannians*, 2025. PhD thesis in Mathematics at Universidad de Cantabria, supervised by Professors Carlos Beltrán and Ujué Etayo. I focused on the classic problem of equidistributing points on spheres, projective spaces, and Grassmannians from two perspectives. On the one hand, we study the Riesz and logarithmic energies of finite collections of points on these spaces. On the other hand, we explicitly construct measure-preserving mappings from the unit cube to spheres and projective spaces of any dimension, among other spaces. These mappings provide an alternative method for generating well-distributed collections of points on these manifolds.
3. *Separation results for minimal energy points on spheres*, 2021. MSc thesis in Mathematics and Computer Science at Universidad de Cantabria, supervised by Professor Carlos Beltrán. I studied separation properties of minimizers of the Riesz and logarithmic energy on spheres.
2. *On the Hopf fibration and the parallelization of spheres*, 2020. BSc thesis in Mathematics at Universidad de Cantabria, supervised by Professor Fernando Etayo. I studied different constructions of the classical Hopf fibration and its generalization to spheres of other dimensions. I also constructed explicit parallelizations and almost complex structures of spheres as well as parallelizations of their product.
1. *From Bloch oscillations to Ohmic transport*, 2019. BSc thesis in Physics at Universidad de Cantabria, supervised by Professor Pablo García-Fernández. I studied the oscillatory behavior of electrons in a perfectly periodic ideal crystal and the evolution of these oscillations into Ohmic transport applying time-dependent second-principles density functional theory on a purely quantum one-dimensional model.

AWARDS AND HONORS

Best poster award in the Congreso Bienal de la Real Sociedad Matemática Española, Ciudad Real (Spain), January 2022.

GRANTS AND SCHOLARSHIPS

Predoctoral grant

2022–2026

Funded by the Spanish government.

Scholarship for university students

2015–2017, 2018–2021

Granted by the Spanish government.

Scholarship for outstanding university students

2014–2015

Granted by Fundación Botín.

RESEARCH STAYS**University of Vienna**, Vienna (Austria)

September 2024–December 2024

Host: Professor Martin Ehler

University of Minnesota, Minneapolis (USA)

April 2023–May 2023

Host: Professor Dmitriy Bilyk

TALKS AT CONFERENCES AND SEMINARS

18. *A new upper bound for the minimal Riesz 2-energy on the sphere*, Advances in Optimal Point Distributions and Mathematical Analysis, Barcelona (Spain), May 2026.
17. *Random point configurations and minimal energy on the sphere and beyond*, Seminar of the Department of Mathematics and Department of Quantitative Methods, CUNEF Universidad, Madrid (Spain), May 2026.
16. *Points on $SO(3)$ with low logarithmic energy*, Congreso Bienal de la Real Sociedad Matemática Española, Alicante (Spain), January 2026.
15. *Minimal Riesz energy on the Grassmannian $Gr_{2,4}$* , International conference: Constructive Functions 2025 in conjunction with the 37th Shanks Lecture, Nashville (USA), May 2025.
14. *Minimización de energía en esferas, espacios proyectivos y grassmannianas*, Universidad de Cantabria, Santander (Spain), May 2025.
13. *Energy minimization on spheres, projective spaces, and Grassmannians*, Seminar of the Department of Mathematics and Department of Quantitative Methods, CUNEF Universidad, Madrid (Spain), February 2025.
12. *Lower and upper bounds for the Riesz energy on the Grassmannian*. VII Congreso de Jóvenes Investigadores de la RSME, Bilbao (Spain), January 2025.
11. *Upper bounds for the Riesz energy on the Grassmannian using determinantal point processes*, Seminar of the Applied Harmonic Analysis Cluster, Vienna (Austria), November 2024.
10. *Well-distributed points on the sphere and the real projective plane*. XXVIII CEDYA / XVIII CMA conference, Bilbao (Spain), June 2024.
9. *Measure-preserving mapping from the unit cube to spheres and projective spaces*. Congreso Bienal de la Real Sociedad Matemática Española, Pamplona (Spain), January 2024.
8. *A family of well-distributed points on the sphere with antipodal symmetry*, Sphere packings, coverings, and spherical codes SPCSC2023, Sofia (Bulgaria), May 2023.
7. *Low-energy points on the sphere and the real projective plane*. International Conference on Approximation Theory and Beyond, Nashville (USA), May 2023.

6. *In search of energy minimizers on the Grassmannian*. VI Congreso de Jóvenes Investigadores de la RSME, León (Spain), February 2023.
5. *Aplicaciones equivolumétricas del hipercubo unidad a las variedades armónicas clásicas*. II Encuentro conjunto RSME-UMA, Ronda (Spain), December 2022.
4. *Towards an optimal discretization of the real projective plane*. BYMAT Conference, online, November 2022.
3. *Minimizing the Green energy on the real projective plane*. Random and Deterministic Point Configurations. Lluís Santaló School 2022, Santander (Spain), July 2022.
2. *Distribución de puntos en la esfera y el plano proyectivo real*. Encuentros de Primavera de Matemáticas en la EII (Málaga)-I, Málaga (Spain), May 2022.
1. *Distribución de puntos en el plano proyectivo real*. Congreso Bienal de la Real Sociedad Matemática Española, Ciudad Real (Spain), January 2022.

POSTER CONTRIBUTIONS

10. *Asymptotically optimal t -design curves on the Grassmann manifold*, 15th International Conference on Sampling Theory and Applications (SampTA), Vienna (Austria), 2025.
9. *Measure-preserving mapping from the unit cube to spheres and projective spaces*. Congreso Bienal de la Real Sociedad Matemática Española, Pamplona (Spain), January 2024.
8. *Measure-preserving mapping from the unit cube to spheres and projective spaces*. Semana de la Ciencia 2023, Santander (Spain), November 2023.
7. *Low-energy points on the sphere and the real projective plane*. Foundations of Computational Mathematics (FoCM) conference, Paris (France), June 2023.
6. *Towards an optimal discretization of the real projective plane*. Recent Advances in Applications of Harmonic Analysis to Convex Geometry, Fargo (USA), April 2023.
5. *The projective Diamond ensemble*. II Encuentro conjunto RSME-UMA, Ronda (Spain), December 2022.
4. *Distributing points on the real projective plane*. Semana de la Ciencia 2022, Santander (Spain), November 2022.
3. *Distributing points on the real projective plane*. 10th International Conference on Curves and Surfaces, Arcachon (France), June 2022.
2. *Resultados de separación para puntos de mínima energía en esferas*. Congreso Bienal de la Real Sociedad Matemática Española, Ciudad Real (Spain), January 2022.
1. *From Bloch Oscillations to Ohmic Transport using Second-Principles Density Functional Theory*. XXIV International Symposium on the Jahn-Teller Effect, Santander (Spain), June 2018.

OTHER ATTENDED WORKSHOPS AND SCHOOLS

4. Rivière-Fabes Symposium on Analysis and PDE, Minneapolis (USA), April 2023.

3. 15th International Conference on Monte Carlo and Quasi-Monte Carlo Methods in Scientific Computing, Linz (Austria), July 2022.
2. Point Configurations: Deformations and Rigidity, LMS Research School, University College London, London (United Kingdom), June-July 2022.
1. Optimal Point Configurations on Manifolds, Vienna (Austria), January 2022. *Online participant.*

TEACHING EXPERIENCE

2025-2026

Statistics. BSc in Economics, 2nd year, CUNEF Universidad. 60 hours.

Mathematics II. BSc in Economics, 1st year, CUNEF Universidad. 60 hours.

2023-2024

Numerical Methods. BSc in Physics, 3rd year, Universidad de Cantabria. 20 hours.

Integral Calculus. BSc in Mathematics, 1st year, Universidad de Cantabria. 20 hours.

2022-2023

Further Differential Calculus. BSc in Mathematics, 2nd year, Universidad de Cantabria. 10 hours.

Integral Calculus. BSc in Mathematics, 1st year, Universidad de Cantabria. 20 hours.

SERVICE

Organizer of the special session “Point configurations: energy minimization, discrepancy, and related topics” in the Congreso Bienal de la Real Sociedad Matemática Española 2026, Alicante (Spain), January 2026.

Organizer of a poster session during the Semana de la Ciencia 2022 in Universidad de Cantabria. Santander (Spain), November 2022.

Volunteer in the Congreso Bienal de la Real Sociedad Matemática Española 2019. Santander (Spain), February 2019.

Organization Committee of the XXIV International Symposium on the Jahn-Teller Effect. Santander (Spain), June 2018.

LANGUAGES

- Spanish: Fluent (Native speaker).
- English: Fluent.
- French: Basic.